

WASP-12b

R-1987

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_150220 · created 2026-08-07T14:48:46+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457073.81406 +/- 0.00014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.045
Transit depth (Rp/Rs)^2	2.4 +/- 1.41 [%]
Orbital Inclination (inc)	86.73 +/- 2.83
Airmass coefficient 1 (a1)	0.55 +/- 0.12
Airmass coefficient 2 (a2)	0.51 +/- 0.13
Scatter in the residuals of the lightcurve fit is	32.77 %
Best Comparison Star	#2 - [330, 52]
Optimal Aperture	3.29
Optimal Annulus	7.61
Transit Duration (day)	0.1331 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.045 vs 0.1178 (deviation 0.83σ)
Duration fit vs published	0.1331 d vs 0.125 d (deviation 1.25σ)
Mid-time O–C	-0.2 min
Depth SNR	3.4
Residual scatter	32.77 %

Light curve

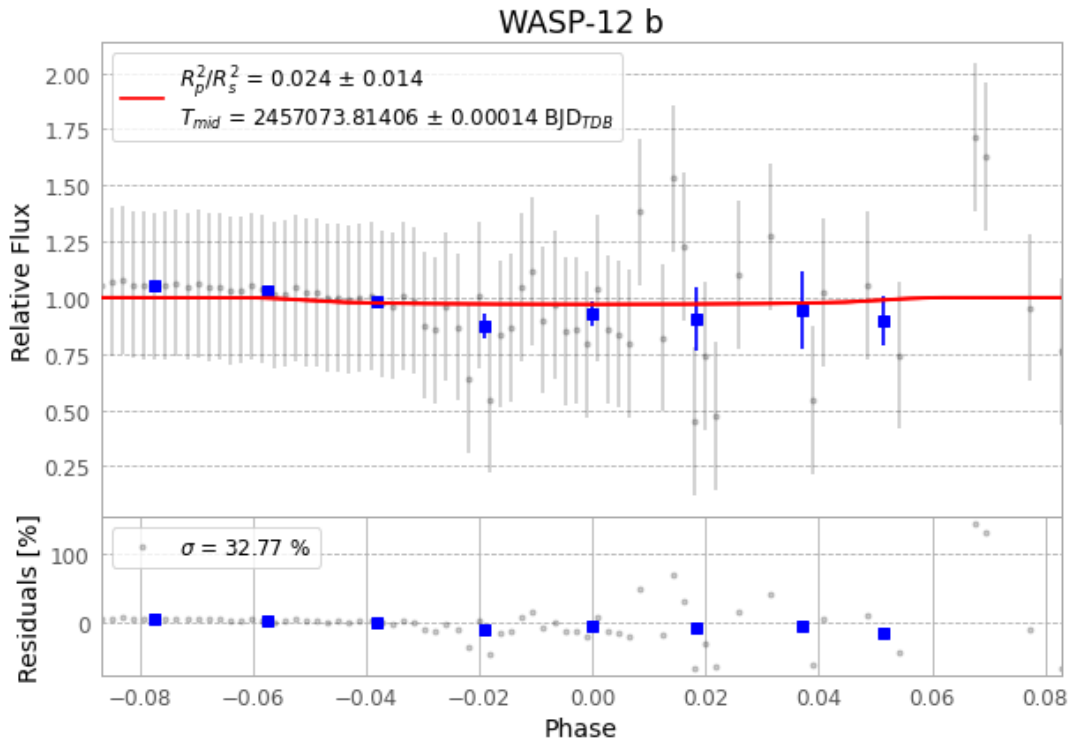


Figure 1 — FinalLightCurve_WASP-12 b_2015-02-19.png (SHA-256 a61f6218d0453eac...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [237, 278], 2 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 4907d102921579bd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

101 FITS frames (66 MB), WASP-12150220050912.FITS → WASP-12150220100912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150220.log (75391e701a9b...), FinalLightCurve_WASP-12 b_2015-02-19.png (a61f6218d045...), FinalLightCurve_WASP-12 b_2015-02-19.csv (bada9c780fad...)

Compute resources

Wall time 180.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 14:48Z.